MINGKE (BRITNEY) TIAN

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EDUCATION

B.S. in Applied Mathematics and Statistics, B.S. in Computer Science **Aug. 2023 - May 2027**
Emory University; Emory College of Arts and Science *GPA: 3.983/4.00*

- **Honor:** Dean's List (2023-2026), Honor thesis Program
- **Awards and Grants:** Emory Wilson Family STEM Undergraduate Research Award (2026), National Science Foundation (NSF) Research Grant (2025), Emory Pathway Global Internship Grant (2025), Mathematical Contest in Modeling (MCM), Honorable Mention (2024-2025), Emory Phi Sigma (2024)
- **Coursework:** Graph Theory (Grad. Level), Analysis Seminar (Grad. Level), Real Analysis I&II, Numerical PDEs, Numerical Analysis, Mathematical Statistics, Numerical Linear Algebra, Regression Analysis, Generative Model, Probabilistic Machine Learning, Computer Vision, Data Structure & Algorithms, Computer Architecture, Analysis of Algorithm, Database Systems, Operating Systems, Foundations of HCI, Data Science Computing, Text Data Science

PUBLICATION

- Zeeshan Memon, ***Mingke Tian,*** Xinyuan Song, Hongwei Jin, Kibaek Kim, Liang Zhao. *CONSTRAINER: Promptable Graph-Structured Optimization via Constraint Conditioning.* ***Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026)***, accepted, to appear. [paper]
- ***Mingke Tian***, Yiyun Chen, Ruipeng Li, Tianshi Xu, Yuanzhe Xi. *ATSolver: Agentic Tuning of Sparse Linear Solvers with Numerical Feedback*. arXiv, 2026. link. [paper]

RESEARCH EXPERIENCE

CONSTRAINER: Promptable Graph-Structured Optimization via Constraint Conditioning **Sep. 2025 - Feb. 2026**
Department of Computer Science & Argonne National Laboratory *Atlanta, GA*

- Proposed CONSTRAINER, a constrained-conditioned foundation model for graph-structured optimization, enabling zero-shot transfer and generalization to power-grid optimization and planning without specific retraining.
- Designed an operator-DAG constraint encoder, preserving constraint semantics through contrastive learning over equivalent rewrites and feasibility-changing constraints, reducing optimality gap by 45% (0.91% to 0.50%).
- Integrated constraint-conditioned cross-attention to a heterogeneous graph transformer, achieved 99.02% feasibility.
- Built scalable training along 16 combinatorial tasks; reached convergence in approx. 30% fewer epochs.

ATSolver: Agentic Tuning of Sparse Linear Solvers with Numerical Feedback **Jul. 2026 - Oct. 2026**
Department of Mathematics & Lawrence Livermore National Laboratory *Atlanta, GA*

- Developed ATSolver, a verifier-grounded agentic tuner LLM reasoning and numerical execution, integrating trusted workload cards, multimodal matrix views, and retrieved experience to proposesolver–preconditioner configurations.
- Designed a LLM policy over typed solver and lifecycle actions, balancing setup cost, solve time, and reuse.
- Built a gated propose–execute–diagnose–repair loop, using feedback to revise parameters or solver branches.
- Achieved 75/105 acceptance under matched-budget Best-of-6 with 30.1% lower replay solve time on 35 matrices
- Retrieval held-out PDE replay target attainment 19/42 to 38/42; cut 32-RHS deployment time 48.5% vs. expert rule.
- First Author manuscript published to arXiv and under review at **ICLR 2027 Main**.

MobilityWorld: Learning a Structured World Model for Human Mobility via Latent State-Action Dynamics **Mar. 2026 - Jul. 2026**
Mingke Tian*, Riyang Bao*, Ruiyu Ning, Dazhou Yu, Andreas Zufle, Liang Zhao *Atlanta, GA*

- Developed a latent world model for sparse GPS, unifying inference, rollout, and graph-executable planning.
- Designed structured consistency learning and differentiable temporal MAP to jointly recover latent state and action.

- Evaluated eight mobility tasks through a unified query interface over shared dynamics
- First Author manuscript under review at **AAAI 2027 Main**.

StayBERT: Staypoint Detection as the Pretext Task for Trajectory Foundation Models **Mar. 2026 - Jul. 2026**
Riyang Bao*, ***Mingke Tian****, Andreas Zufle, Liang Zhao *Atlanta, GA*

- Developed StayBERT, a trajectory foundation model pretrained through supervised staypoint detection on raw GPS.
- Fused Transformer motion and place features via dimension-wise gate; decoded coherent staypoint spans with CRF.
- Evaluated frozen features on seven tasks across three datasets; achieved 89.8% activity F1 in cross-city transfer.
- Co-First Author manuscript under review at **AAAI 2027 Main**.

Structured MAGI with Warm Starts for Partially Observed Bayesian ODE Inverse Problem **Jun. 2025 - Dec. 2025**
Tianshi Xu, ***Mingke Tian***, Yuanzhe Xi 、 *Atlanta, GA*

- Developed physics-informed Bayesian inference for partially observed ODEs, jointly estimating latent trajectories and parameters from sparse and noisy observations with uncertainty quantification.
- Combined multi-task GP denoising with staged PINN estimates to initialize HMC and reduce burn-in steps.
- Analyzed Matérn kernel matrices for HSS structure and numerical rank to support Gaussian process inference.

WORK EXPERIENCE

Siasun Robotics & Automation Co., Ltd. **Jun 2026 - Aug 2026**
Robotics ML Intern *Shenyang, CN*

- Trained and deployed NVIDIA Isaac GR00T N1.7 for real-robot parcel manipulation with PyTorch and demos.
- Mapped normalized 16-step chunks to SIASUN gripper control; verified framing, VRAM, and concurrency.
- Switched absolute EEF to relative deltas after ~22% success, cutting offline position MAE from 18.31 to 5.78 cm.
- Aggregated SpaceMouse corrections at policy failures for DAgger-style post-training and real-robot policy tests.
- Robot grasp-and-lift rollouts raised autonomous success from ~30% to 55% and cut intervention by ~25%.

Ricoh USA **Sep 2025 - Dec 2025**
AI/Machine Learning Engineer Intern, Co-op *Atlanta, GA*

- Built Qwen2.5-VL-7B extraction on SageMaker, fusing OCR/layout for ~600 enterprise documents (3.2K pages).
- LoRA-tuned labeled page groups, raising F1 from 86.7% to 92.3% and schema validity from 93.4% to 98.6%.
- Raised cross-page field accuracy from 78.9% to 88.7% and improved citation precision via retrieval and attribution.
- Built an MLflow evaluation pipeline; prompt caching cut processing time 20.2%, with F1 within 0.3 points of peak.

InnovaAI Tech LLC **Jun 2025 - Aug 2025**
AI Engineer Intern *San Francisco, CA*

- Deployed LangChain agents via FastAPI/Docker with routing, planning, persona control, and tool calling.
- Built FAISS/Chroma RAG with adaptive semantic chunking, retrieval, reranking, and source-grounded generation.
- Integrated Slack and Google Calendar/Tasks, handling ~78% of routine requests and cutting scheduling time ~65%.
- Streamed and async concurrency delivered p95 time-to-first-token <900 ms with <0.6% tool errors under rate limits.

LEADERSHIP, PROFESSIONAL DEVELOPMENT & OUTREACH

Emory University Mathematics Association (EUMMA) **Atlanta, GA**
President *Aug. 2024 – Present*

- Led 100+ members and launched "AI4SE" initiative to promote AI literacy through workshops, and mentorship.
- Organized research and networking panels; Led monthly technical training for Kaggle and other math competitions.

Emory Student Government Association *First Year Councilor*
MIT AwesomeMath Summer Session, Algebra & Geometry *Teaching Assistant*
Atlanta Mathematics of Data Science Bootcamp *Student Volunteer & Attendee*
Atlanta Science Festival, Exploration Exposition *Student Volunteer*